Hilbert Space Comparison: FFGFT $L^2(T^4)$ and UMF $\ell^2(P_N)$

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Abstract

Marco Gericke (Universal Model Framework, UMF) has proposed a formal architecture in which quantum mechanics, special relativity, and fermion mass emerge from a prime-indexed Hilbert space $\ell^2(P_N)$ and its reciprocal interior $\ell^2(P_N^{-1})$, coupled by a bivector B_{OI}. This document analyses the structural relationship between UMF and FFGFT. The central result is: FFGFT's $L^2(T^4)$ is the more complete Hilbert space — it contains UMF's $\ell^2(P_N)$ as a proper subspace. However, UMF's prime-indexed structure offers genuine computational advantages in specific domains, particularly prime factorisation and number-theoretic problems. The two frameworks

are therefore not in competition but stand in a relation of **mathematical containment (subset relation) and physical complementarity**: UMF's Hilbert space is a proper subspace of FFGFT's, but UMF's prime-indexed structure, bivector B_{OI}, and activation operator K_{Kriger} provide formal tools that FFGFT currently lacks.

Notation

- $P_N = \{p_1, p_2, \ldots, p_N\}$ with $p_1 = 2, p_2 = 3, p_3 = 5, \ldots$ (first N primes)
- $T^4 = (\mathbb{R}/2\pi\mathbb{Z})^4$ (4-torus, FFGFT state space)
- $B_{OI} = \Pi_{\text{out}} \wedge \Pi_{\text{in}} \in \Lambda^2(H_{\text{out}} \oplus H_{\text{in}})$ (UMF bivector, see §.2)
- $\xi = 4/30000$ (FFGFT dimensionless winding parameter)
- K_{Kriger}: UMF activation operator (see §.3)
- ξ_{UIFT}: dimensionless parameter of the UIFT framework (Doc. 013), related to ξ_{FFGFT} via the bridge factor $414{,}684$

List of Symbols

Symbol	Meaning
$\xi = 4/30000$	FFGFT dimensionless winding parameter (geometrically fixed, $\kappa = 7$)
ξ_{FFGFT}	Same as ξ; used when distinguishing from ξ_{UIFT}
ξ_{UIFT}	Dimensionless parameter of UIFT: $k_B T_{\text{CMB}} \ln 2/(m_e c^2)$ (Doc. 013)
$L^2(T^4)$	Hilbert space of square-integrable functions on the 4-torus T^4
T^4	4-dimensional torus $(\mathbb{R}/2\pi\mathbb{Z})^4$; FFGFT state space
$\ell^2(P_N)$	UMF Hilbert space: square-summable sequences over first N primes
P_N	Set of first N prime numbers $\{2, 3, 5, \ldots, p_N\}$
$w_p = \ln p/\sqrt{p}$	Prime weight in UMF inner product
B_{OI}	UMF bivector: $\Pi_{\text{out}} \wedge \Pi_{\text{in}}$; couples the two spaces
Π_{out}	UMF Prime Boundary $= \ell^2(P_N)$ (exterior, metric domain)
Π_{in}	UMF Reciprocal Interior $= \ell^2(P_N^{-1}) \oplus \ell^2(Q_{ZN})$ (mass domain)
K_{Kriger}	UMF activation operator; determines which interface information generates curvature
$\chi(x)$	Dimensionless activation parameter: $k_B T(x) \ln 2 \cdot \tau_t(x)/(\hbar/4)$
$\chi = 1$	Landauer threshold: boundary between inactive and active structure
D_P	UMF chiral Dirac operator on $\ell^2(P_N) \otimes \mathbb{C}^2$
$\tau \cdot \mu$	Coherence-time – information-mass product; $= \hbar/c^2$ (both frameworks)
$\varepsilon \approx 0.001$	FFGFT non-closure of T^4; generative structural residual
θ_4	Continuous fourth-torus phase coordinate; absent from $\ell^2(P_N)$
n_θ, n_ϕ	FFGFT winding numbers in two independent torus directions
ι	Embedding $\ell^2(P_N) \hookrightarrow L^2(T^4)$ (containment map)

.1 The Two Hilbert Spaces

FFGFT: $L^2(T^4)$

FFGFT's state space is $L^2(T^4)$ — the space of square-integrable functions on the 4D torus $T^4 = (\mathbb{R}/2\pi\mathbb{Z})^4$. The natural orthonormal basis consists of Fourier modes:

$$e_{\mathbf{n}}(\boldsymbol{\theta}) = \frac{1}{(2\pi)^2} e^{i(n_1\theta_1+n_2\theta_2+n_3\theta_3+n_4\theta_4)}, \qquad \mathbf{n} \in \mathbb{Z}^4 \tag{244.1}$$

This space is:

- **Complete**: every Cauchy sequence converges in $L^2(T^4)$
- **Separable**: the Fourier basis $\{e_{\mathbf{n}}\}_{\mathbf{n}\in\mathbb{Z}^4}$ is countable and dense
- **Continuous**: admits continuous winding numbers and phase structure
- **Indexed by** $\mathbb{Z}^4$: all integer winding number combinations, including prime-indexed ones as a special case

The single parameter $\xi = 4/30000$ determines which winding modes are dynamically preferred — but all modes are structurally present in $L^2(T^4)$.

UMF: $\ell^2(P_N)$

UMF's primary space is $\ell^2(P_N)$ — the space of square-summable sequences indexed by the first N prime numbers, with inner product:

$$\langle f|g\rangle_{P_N} = \sum_{p\leq p_N} w_p\,\overline{f(p)}\,g(p), \qquad w_p = \frac{\ln p}{\sqrt{p}} \tag{244.2}$$

This space is:

- **Complete** for fixed finite N: $\ell^2(P_N) \cong \mathbb{C}^N$
- **Discrete**: no continuous phase structure
- **Indexed by primes only**: $\{2, 3, 5, 7, 11, \dots, p_N\}$
- **Weight-divergent** as $N \to \infty$: $\sum_p w_p = \sum_p \ln p/\sqrt{p}$ diverges (by comparison with $\sum 1/\sqrt{p}$)

.2 Containment Relation

$\ell^2(P_N)$ is a Subspace of $L^2(T^4)$

The prime-indexed basis vectors of UMF correspond to specific Fourier modes of $L^2(T^4)$. Consider the embedding:

$$\iota : \ell^2(P_N) \hookrightarrow L^2(T^4), \qquad f(p) \mapsto f(p) \cdot e_{(p,0,0,0)}(\boldsymbol{\theta}) \tag{244.3}$$

This maps each prime-indexed coefficient to a Fourier mode with winding number $(p, 0, 0, 0)$ in the first torus direction. The embedding is isometric up to normalisation and injective. Therefore:

$$\boxed{\ell^2(P_N) \hookrightarrow L^2(T^4), \quad \text{proper subspace}} \tag{244.4}$$

$L^2(T^4)$ contains far more: all integer winding numbers in all four directions, all non-prime modes, all mixed-direction modes, and the continuous phase structure θ_4 which is absent from $\ell^2(P_N)$ entirely.

What UMF's Space Cannot Represent

The following FFGFT structures have no counterpart in $\ell^2(P_N)$:

FFGFT Structure	In $L^2(T^4)$	In $\ell^2(P_N)$
Continuous phase θ_4	(full range)	× (absent)
Non-prime winding modes		×
Mixed-direction winding (n_1, n_2, n_3, n_4)		×
Non-closure ε (winding residual)		×
Z3³ topological constraint		partially
4π spinor periodicity		×
Lorentz structure (from T^4 geometry)		derived via RG

The Missing Object: UMF Bivector B_{OI}

Marco Gericke's central formal contribution is not $\ell^2(P_N)$ alone but the wedge product of the two spaces:

$$B_{OI} = \Pi_{\text{out}} \wedge \Pi_{\text{in}} \in \Lambda^2(H_{\text{out}} \oplus H_{\text{in}}) \tag{244.4b}$$

where $\Pi_{\text{out}} = \ell^2(P_N)$ and $\Pi_{\text{in}} = \ell^2(P_N^{-1}) \oplus \ell^2(Q_{ZN})$.

B_{OI} has three crucial properties:

1. **Projection-stable**: neither π_1 (metric projection) nor π_2 (spinor/-mass projection) alone contains it, but both projections preserve its existence as a structural invariant.
2. **Generates observable geometry**: $g_{\mu\nu}^{\text{phys}} = g_{\mu\nu}^{(0)} + \lambda_B B_{\mu\alpha}^{\text{act}} B_\nu^{\text{act}\,\alpha}$
3. **Not derivable from either space alone**: it is the coupling between Π_{out} and Π_{in}, not a state within either.

This is the formal realisation of what Stefaan Vossen called the GIA projection-stable invariant, and what FFGFT described verbally as "mutual constitution". B_{OI} is *not* an element of $L^2(T^4)$ — it is a relation *between* spaces. The containment $\ell^2(P_N) \hookrightarrow L^2(T^4)$ holds for the spaces themselves; B_{OI} is a genuinely new object that FFGFT currently has no algebraic counterpart for.

.3 Where UMF Offers Genuine Added Value

The containment relation does not make UMF redundant. Three domains where the prime-indexed structure offers advantages:

Prime Factorisation and Number Theory (Doc 075, 076)

FFGFT Doc 075 treats RSA factorisation via T0-field dynamics — period-finding as wave propagation rather than quantum superposition. The natural computational basis for this problem is prime-indexed. UMF's $\ell^2(P_N)$ with the prime-weighted inner product $w_p = \ln p/\sqrt{p}$ is the *natural* Hilbert space for number-theoretic computations: the weights encode the prime number theorem density $\pi(x) \sim x/\ln x$ directly.

In $L^2(T^4)$, prime modes are present but not distinguished. In $\ell^2(P_N)$, they are the entire basis. For factorisation problems, UMF's representation is computationally more efficient.

The Dirac Operator D_P as Computable Spectrum

UMF's chiral Dirac operator on $\ell^2(P_N) \otimes \mathbb{C}^2$:

$$D_P = \begin{pmatrix} 0 & A \\ A^\dagger & 0 \end{pmatrix}, \quad A_{pq} = w_p \, \delta_{pq} \tag{244.5}$$

has a discrete, explicitly computable spectrum. The spectrum of the corresponding operator on $L^2(T^4)$ requires solving the full torus eigenvalue problem — substantially harder. For mass-spectrum calculations in the lepton sector, both approaches give $\sim 1\%$ agreement with PDG, but UMF's computation is more direct.

Independent Confirmation of Lorentz Invariance

UMF derives Lorentz invariance as the IR fixed point of the anisotropy RG flow $u(\mu) \to u^* = 1$. FFGFT derives it from the T^4 winding geometry and the constraint $\tilde{T} \cdot m = 1$. These are structurally independent derivations arriving at the same result. The independence strengthens both: if two different mathematical starting points require Lorentz invariance, the result is more robust.

The Activation Operator K_{Kriger}

UMF defines the dimensionless activation parameter:

$$\chi(x) = \frac{k_B T(x) \ln 2 \cdot \tau_t(x)}{\hbar/4} \tag{244.6a}$$

and the soft activation gate:

$$A(x) = \frac{1}{1 + e^{-s(\chi(x)-1)}} \tag{244.6b}$$

Physical geometry receives contributions only from the activated fraction:

$$\Phi_{\text{grav}} = A(x) \cdot \Phi_{\text{interface}} \tag{244.6c}$$

Key insight: The threshold $\chi = 1$ is exactly the Landauer bound. For $\chi < 1$, the bivector B_{OI} exists mathematically but generates no curvature — it is structurally present but physically inactive. This formalises what FFGFT has described qualitatively as the boundary between geometrically present and observationally active structure. FFGFT has no algebraic analogue of K_{Kriger}. It is UMF's genuine formal contribution, independent of the containment relation.

.4 The $\tau \cdot \mu$ Identity: Independent Derivation

Both frameworks arrive at the same identity independently:

$$\tau \cdot \mu = \frac{\hbar}{c^2} \tag{244.7}$$

where $\tau = \hbar/(k_B T_{\text{CMB}} \ln 2)$ and $\mu = k_B T_{\text{CMB}} \ln 2/c^2$.

In FFGFT this is the coherence time – information-mass product at the CMB scale. In UMF this is the statement that the Kriger activation threshold, evaluated at T_{CMB}, equals the Compton-scale information mass. Two different derivational paths, identical result. The agreement is not coincidental: it reflects the bridge formula $\xi_{\text{FFGFT}}/\xi_{\text{UIFT}} \approx 414{,}684$ (Doc 190 R13), which connects the same CMB temperature to both frameworks through different normalisations.

.5 The Non-Redundancy Principle

The most useful framework is not the most complete one but the one that derives the most from the fewest assumptions. By this criterion:

Table 1 — Technical comparison:

Criterion	**FFGFT**	**UMF**
Free parameters	1 (ξ)	5 (Distinction Logic axioms + N)
Hilbert space	$L^2(T^4)$ (complete)	$\ell^2(P_N) \subset L^2(T^4)$
Lorentz derivation	from T^4 geometry	from RG flow (independent)
Mass spectrum	from winding recursion	from D_P spectrum (compact)
Activation boundary	qualitative	K_{Kriger} (explicit)
Factorisation	wave propagation	natural prime basis

Table 2 — Ontological / structural comparison:

Criterion	**FFGFT**	**UMF**
Primitive	Geometry (T^4)	Distinction Logic
Bivector B_{OI}	not present	$\Pi_{\text{out}} \wedge \Pi_{\text{in}}$
"Mutual constitution"	described verbally	algebraically realised
Ontological commitment	geometric primacy	distinction-logic primacy

The two frameworks occupy different positions in the projection space $\mathscr{M}$ (Stefaan Vossen's comparative architecture): they share projection-stable invariants (Lorentz structure, $\tau \cdot \mu$, mass spectrum) while differing in generative ontology and computational representation.

.6 Conclusion: Complementarity without Redundancy

The analysis yields five results that are independent of any particular exchange or context and follow directly from the structural comparison above.

1. **Containment.** $\ell^2(P_N) \hookrightarrow L^2(T^4)$: UMF's Hilbert space is a proper subspace of FFGFT's. $L^2(T^4)$ is the structurally more complete space. This is a mathematical fact, not a value judgement about either framework.
2. **Non-redundancy.** The containment relation does not make UMF redundant. UMF offers computational advantages in prime-indexed domains, an explicit activation operator (K_{Kriger}), an independent derivation of Lorentz invariance via RG flow, and a computable Dirac spectrum. These are genuine contributions not present in $L^2(T^4)$.
3. **Structural alignment.** The $\tau \cdot \mu$ identity $= \hbar/c^2$ is derived independently in both frameworks and confirms structural convergence at the CMB scale (Doc. 013).
4. **The bivector as a new object.** UMF's $B_{OI} = \Pi_{\text{out}} \wedge \Pi_{\text{in}}$ and the activation operator K_{Kriger} are formal objects FFGFT currently lacks. They are relations *between* Hilbert spaces, not states within one space, and are therefore not contained in $L^2(T^4)$. A natural next step is to map ξ_{FFGFT} to the UMF compactification radius R_χ via the bridge factor $\xi_{\text{FFGFT}}/\xi_{\text{UIFT}} \approx 414{,}684$ (Doc. 013), and to complete the UMF entry in the comparative projection schema (Vossen 2026).
5. **Ontological neutrality.** No ontological priority claim is required by this analysis. UMF derives $\ell^2(P_N)$ from distinction logic; FFGFT derives $L^2(T^4)$ from geometry. The subspace relation holds regardless of which generative account one adopts.